

JEE MAIN REVISION MASTER: AREA UNDER THE CURVES 2026 1 QUESTIONS 1.67% WEIGHTAGE

↓49.85%

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 17 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Area Under The Curves - Core Application	17 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Area Under The Curves - Core Application (Very High Yield — 17 PYQs)

Core Principle: Key problem pattern in Kcet Area Under The Curves.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The area of the region bounded by the curve $y^2=x^3$, the y-axis and the lines $y=1$ and $y=8$ is..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "The area bounded by the curve $y=\sin((x/3))$, x-axis, the lines $x=0$ and $x=3\pi$ is..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "The area of the region bounded by the curve $y=x^2$ and the line $y=16$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The area of the region bounded by the curve $y^2 = x^3$, the y -axis and the lines $y=1$ and $y=8$ is

- (A) $(155 / 3)$ sq. units
- (B) $(93 / 5)$ sq. units
- (C) 93 sq. units
- (D) 155 sq. units

Correct Answer: None

Solution Analysis: The curve is $y^2 = x^3$. Since the region is between the lines $y=1$ and $y=8$, it is easier to write x in terms of y . From $y^2 = x^3$ we get $x = y^{2/3}$. The y -axis is $x=0$. So, for each value of y from 1 to 8, the horizontal width of the region is $x_{\text{right}} - x_{\text{left}} = y^{2/3} - 0 = y^{2/3}$. Hence, the required area is $\int_1^8 y^{2/3} dy$. Now, $\int y^{2/3} dy = \frac{y^{5/3}}{5/3} = \frac{3}{5} y^{5/3}$. So, Area = $\left[\frac{3}{5} y^{5/3} \right]_1^8 = \frac{3}{5} (8^{5/3} - 1^{5/3})$. Since $8^{1/3} = 2$ we have $8^{5/3} = 2^5 = 32$. Thus, Area = $\frac{3}{5} (32 - 1)$.